

Research Article

Observed precipitation extremes and near term CMIP6 projections for Uttaradit

Thailand

[Author names and affiliations to be supplied]

[Corresponding author and email to be supplied]

Research draft requiring confirmation of data provenance before submission

Abstract

This study evaluates annual precipitation extremes at 13 stations in Uttaradit, Thailand, using observations for 1981-2014 and seven supplied CMIP6 model series under SSP2-4.5 and SSP5-8.5 for 2021-2050. Eleven indices describe wet-day totals, rainfall intensity, extreme accumulations, threshold exceedances and dry or wet spells. Station-specific observed wet-day percentile thresholds were held fixed across model periods. Mann-Kendall and detrended effective-sample-size modified Mann-Kendall tests were applied alongside Sen slopes. Future mean changes were calculated against each model's 1995-2014 historical baseline. The supplied dataset yielded 143 observed and 2,002 future station-index trend estimates. Modified-test classifications identified 66 observed significant trends, compared with 30 from the standard test; this sensitivity requires caution when interpreting individual p-values. Across stations, the median of seven-model station medians indicated PRCPTOT increases of approximately 2.5% and 1.9% under SSP2-4.5 and SSP5-8.5, respectively. Corresponding Rx1day increases were 6.6% and 8.6%, and R50mm increases were 24.2% and 16.7%. These descriptive changes suggest that increases in some heavy-rainfall measures can accompany small changes in total wet-day precipitation. Spatial interpolation illustrates station-referenced patterns. The findings remain conditional on the supplied preprocessing, annual calendar treatment and incomplete information on gauge homogeneity and bias correction.

Keywords: Precipitation extremes, Uttaradit, CMIP6, Mann-Kendall, Sen slope, Climate projections

1. Introduction

Annual rainfall totals alone cannot describe the intensity, timing and persistence of precipitation extremes. Standardized indices based on daily observations provide complementary information on heavy rainfall and dry or wet spells [1,2]. For engineering applications, distinguishing these measures helps frame questions about drainage, storage and drought exposure. An index change does not itself quantify infrastructure failure or flood discharge, which require additional hydrological and design information.

CMIP6 provides coordinated climate simulations for examining historical conditions and future scenarios [3,4]. Local interpretation nevertheless depends on the model realization, spatial sampling and preprocessing of the supplied series. Trend tests also depend on serial correlation. Yue and Wang [5] showed that effective sample size estimated after detrending can reduce the influence of serial correlation on Mann-Kendall inference. This study applies that framework to the available Uttaradit station series and keeps standard-test results visible for comparison.

The objectives are to characterize observed annual precipitation extremes, quantify model-consistent near-term changes under two scenarios, and document a reproducible implementation. The analysis distinguishes change in the 2021-2050 mean relative to 1995-2014 from monotonic trend within 2021-2050. These are separate quantities and need not have the same sign.

2. Materials and methods

2.1 Study network and supplied data

The network comprises stations 351001-351012 and 351201. Coordinates were read from the supplied station workbook. Observed records cover 1981-2014. Twenty-one supplied

precipitation CSV files represent historical, SSP2-4.5 and SSP5-8.5 data from ACCESS-ESM1-5, CanESM5, CESM2, EC-Earth3, FGOALS-g3, MIROC6 and MRI-ESM2-0. The analysis uses the 13 target station columns. The filenames designate the model series as bias-corrected; the original correction algorithm, training period, validation and observational agency were not established from the files inspected. These details must be supplied and checked before submission. Rainfall values are treated as daily millimetres, an input assumption requiring confirmation from the data provider.

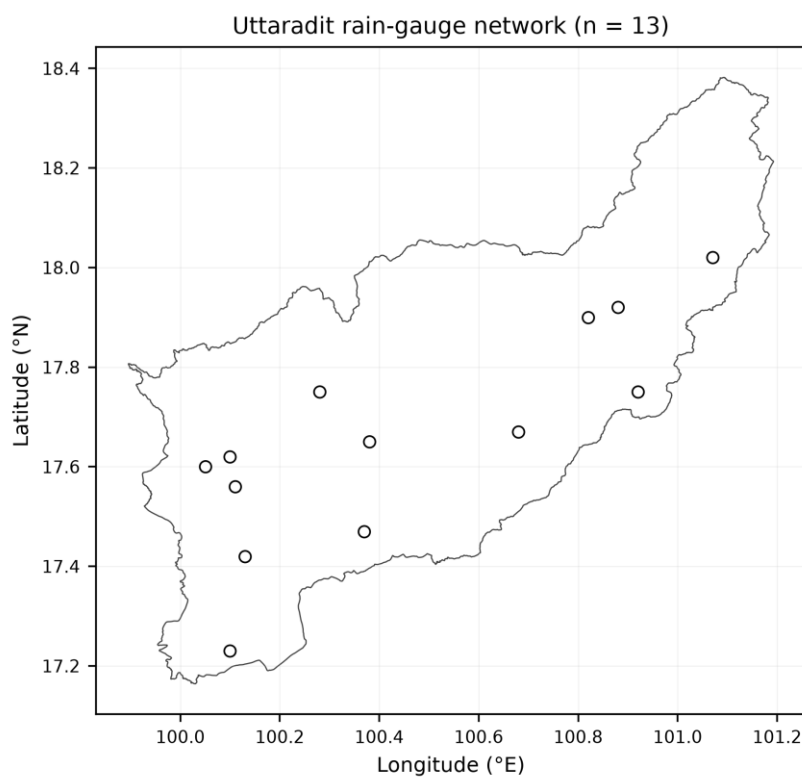


Figure 1 Locations of the 13 supplied stations and the Uttaradit polygon selected by province code 53 from the supplied GIS file.

2.2 Quality control and annual indices

Dates, station identifiers and coordinates were validated before calculation. Duplicate dates, negative rainfall and non-numeric rainfall are rejected. Each station-year is aligned to a Gregorian daily calendar. At least 90% of days must contain valid rainfall; otherwise all annual indices are missing. An unrecorded day breaks a run and prevents a five-day accumulation

spanning that gap. This is a study-specific annual completeness policy rather than a claim of full ETCCDI software equivalence. Some model files omit leap days; their native calendar and any calendar conversion need confirmation. No missing day is imputed.

Wet days satisfy $RR \geq 1$ mm. PRCPTOT sums wet-day precipitation and SDII divides that sum by the number of wet days. Rx1day is the annual daily maximum; Rx5day is the maximum complete five-day running sum within the calendar year. CDD and CWD are the longest within-year runs below and at or above 1 mm, respectively. Spells and Rx5day windows are bounded by calendar years in this implementation. R10mm, R20mm and R50mm count days with rainfall at or above the stated threshold; R50mm is the selected user-defined threshold index. R95p and R99p sum rainfall strictly exceeding observed station-specific wet-day percentiles [1,2].

Percentiles use the full observed 1981-2014 reference sample and the NumPy linear quantile convention. The same thresholds are used for model historical and future series. No in-base-period bootstrap is applied; the resulting percentile indices are therefore described as fixed-reference ETCCDI-based indices. Threshold uncertainty is not propagated. The threshold CSV preserves the reference values and period.

2.3 Trend estimation

The Mann-Kendall statistic S sums the signs of all later-minus-earlier annual differences. Its variance includes the standard correction for tied values. A continuity-corrected normal statistic yields a two-sided p -value. Constant series return $S = 0$, $Z = 0$ and $p = 1$. At least ten valid annual values are required. Sen slope is the median of pairwise changes divided by actual differences in years [6]. A conventional 95% order-statistic slope interval is calculated with SciPy Theil-Sen routines; this interval is not adjusted for serial dependence and is not used as a substitute for modified-test inference.

For the modified test [5], the Sen trend is removed before computing sample autocorrelations with a common mean and full-series sum-of-squares denominator. The variance multiplier is $f = 1 + 2 \sum[(1 - k/n) \rho(k)]$, for lags $k = 1, \dots, n - 1$. Modified variance is $f \text{Var}(S)$, and effective sample size is n/f . All lags are included as specified in the retained analysis design. Factors below one are recorded explicitly and can increase apparent significance. Non-positive factors or irregular retained annual spacing invalidate modified inference. The full-lag estimator is sensitive in short records; results are interpreted alongside standard MK and effect sizes. Individual $\alpha = 0.05$ tests are not multiplicity-adjusted and do not establish regional field significance.

2.4 Future changes and spatial presentation

For each model, station and index, absolute change is the 2021-2050 annual mean minus the 1995-2014 historical annual mean. Relative change divides this difference by the historical mean and multiplies by 100; zero baselines do not produce finite percentages. Descriptive summary values first take the seven-model median at each station, then the median across the 13 stations. This order is retained in Figure 3 and Table 2. It is not equivalent to pooling all 91 model-station values.

Inverse-distance weighting with power two and great-circle distances is applied to station medians of absolute change [7]. Maps are masked to the explicitly identified Uttaradit polygon and displayed only over the calculated station-domain grid. Blank portions have no plotted estimate. Interpolation has not been cross-validated and provides spatial visualization rather than a validated gridded hazard estimate. Model agreement is descriptive; medians of p-values are not ensemble tests.

3. Results

3.1 Data coverage and observed trends

All 442 observed station-years and 5,460 future station-model-scenario-years met the implemented annual completeness criterion. This does not imply every day was present. The output contains 4,862 observed annual index values and 60,060 future values. Observed trend series have 34 annual values and future trend series have 30. The separate output checker confirmed expected dimensions, bounded p-values, ordered slope intervals and model-consistent PRCPTOT means recomputed from raw daily input.

Observed modified-test classifications were spatially mixed (Figure 2). Of 143 tests, 66 were significant under the implemented modified test, compared with 30 under standard MK. For PRCPTOT, two stations increased, two decreased and nine had no significant modified-test trend. R50mm increased at six stations and decreased at two, whereas SDII decreased at six and increased at one. These counts refer to individual tests. They do not demonstrate a province-wide trend after accounting for multiplicity or dependence.

Table 1 Observed modified-test direction counts across the 13 stations, 1981-2014.

Index	Increasing	Decreasing	Not significant
PRCPTOT	2	2	9
SDII	1	6	6
Rx1day	4	1	8
Rx5day	4	3	6
CDD	3	6	4
CWD	2	2	9
R10mm	2	3	8
R20mm	3	5	5
R50mm	6	2	5
R95p	2	3	8
R99p	3	1	9

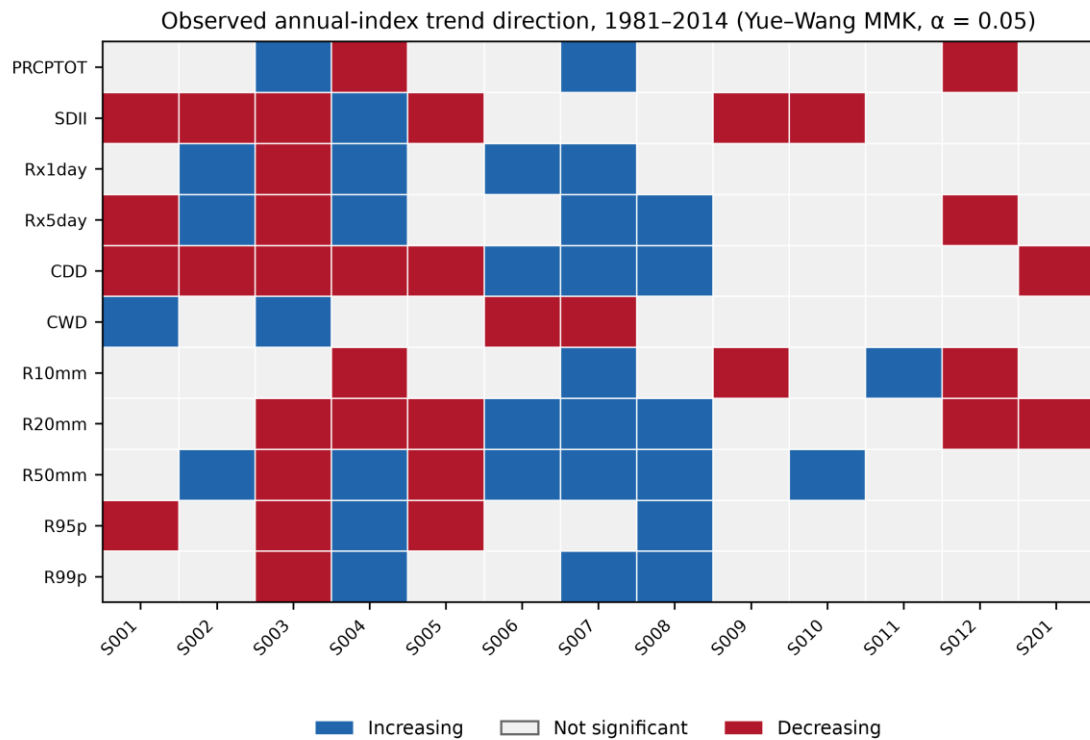


Figure 2 Individual observed station-index trend classifications from the full-lag Yue-Wang implementation. S001-S012 and S201 denote the last three digits of the corresponding station IDs. Significance is unadjusted for multiple testing.

3.2 Projected mean changes and within-window trends

The station-first descriptive summaries indicate small positive PRCPTOT changes under both scenarios, while several heavy-rainfall measures show larger relative increases (Table 2; Figure 3). Rx1day rises by approximately 6.6% and 8.6%, and R50mm by 24.2% and 16.7%, under SSP2-4.5 and SSP5-8.5, respectively. R10mm and R20mm decrease slightly in both summaries. Large percentage changes in infrequent exceedances need to be read with their absolute baseline frequencies.

Table 2 Median relative change (%) for 2021-2050 against model-consistent 1995-2014 baselines. Seven-model medians are calculated per station before taking the median over stations.

Index	SSP2-4.5	SSP5-8.5
PRCPTOT	2.47	1.93
SDII	0.71	-1.52
Rx1day	6.62	8.63
Rx5day	4.27	3.70
CDD	-0.49	-0.97
CWD	1.47	2.65
R10mm	-1.11	-1.51
R20mm	-2.29	-2.29
R50mm	24.24	16.67
R95p	4.45	4.26
R99p	29.22	25.09

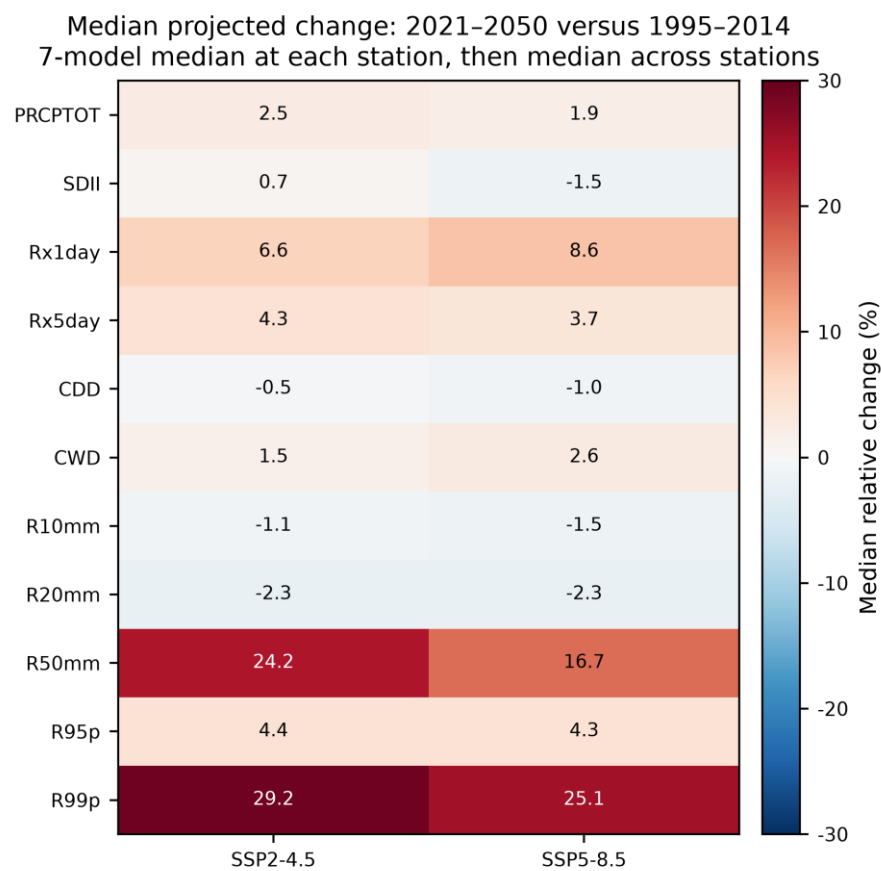


Figure 3 Descriptive changes with the same station-first aggregation as Table 2. Colours represent percentages, not significance.

Within the 2021–2050 window, 445 of 1,001 tests under SSP2-4.5 and 390 of 1,001 under SSP5-8.5 were significant under the full-lag modified test. Decreasing classifications outnumbered increasing classifications in both scenarios (260 versus 185 and 223 versus 167). These within-window tendencies can coexist with a future mean above the earlier baseline.

Figure 4 illustrates station-median PRCPTOT changes and spatial differences between scenarios without assigning significance to interpolated cells.

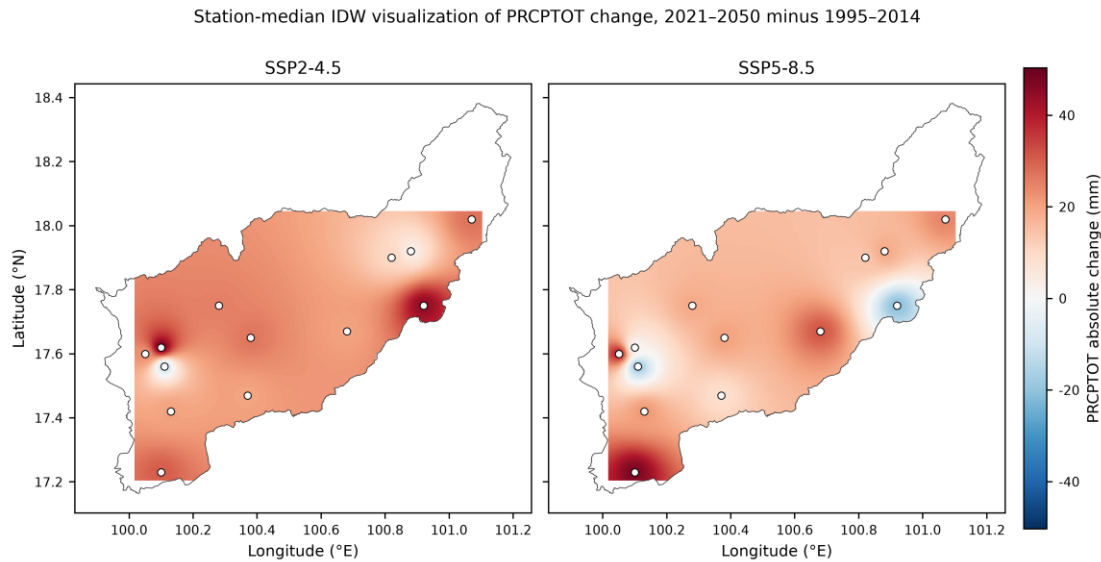


Figure 4 IDW of seven-model station medians of absolute PRCPTOT change. Both panels share a colour scale. White areas outside the displayed station-domain estimate are not zero change. The polygon mask is Uttarakhand only.

4. Discussion

The descriptive projection pattern separates rainfall accumulation from heavy-event metrics. A small increase in PRCPTOT alongside larger Rx1day or threshold-exceedance changes can motivate separate assessment of water supply and intense rainfall exposure. It cannot determine drainage capacity, return-period rainfall or flood damage from these indices alone. Likewise, CDD is a precipitation-spell measure and does not incorporate evapotranspiration or soil water storage.

The discrepancy between standard and modified test classifications is a substantive limitation. The full-lag multiplier falls below one for 128 of 143 observed series and 1,992 of 2,002 future series; consequently modified inference often becomes more permissive. This behaviour must not be described as automatically more conservative or more reliable. A sensitivity analysis of lag treatment and empirical false-positive behaviour would strengthen

inferential claims, particularly with only 30 or 34 annual values. Such analysis has not been completed here. The reported conventional Sen intervals retain their independence assumptions.

Additional limits arise upstream. Data-provider documentation is needed to establish observation quality, rainfall units, model calendars and bias-correction training and validation. A filename alone does not verify preprocessing performance. Fixed observed percentiles aid comparability but omit threshold-estimation uncertainty and differ from bootstrap treatments used in some ETCCDI implementations. Within-year spell and running-window conventions can also alter values near year boundaries. The analysis is consequently a reproducible assessment of the supplied series, pending provenance verification, rather than a validated engineering design standard.

5. Conclusions

The supplied network shows mixed observed trends and descriptive near-term increases in several heavy-rainfall measures. Station-first median PRCPTOT changes are small and positive, whereas Rx1day and R50mm changes are larger in percentage terms. Results depend materially on the inference method: the full-lag modified test yields many more significant classifications than standard MK. Engineering interpretation should emphasize effect sizes, model spread and verified input provenance before adopting these results for design decisions.

6. Acknowledgements and declarations

[To be completed by the authors: data-provider acknowledgement, funding, conflicts of interest, author contributions and any required AI-use disclosure. No declarations have been inferred.]

Data and code availability: The revised code, regression tests, input-file manifest, QC tables and result workbook accompany this draft locally. [Authors must supply data-use permissions and a public repository or justified access statement.]

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